

Mark Verma

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SUMMARY

Machine Learning Engineer with 2 years building production ML/LLM systems in healthcare — MLOps pipelines, distributed data at 500 GB+ scale, and inference optimization (vLLM/SGLang, Hugging Face Endpoints). First-author IEEE paper in applied deep learning.

EXPERIENCE

314e Corp

Data Science Engineer

Bengaluru, India

Jul 2024 – Present

- Built and owned the **MLOps pipeline** for a clinical patient entity-extraction model serving multiple healthcare clients, raising accuracy from **72% to 94%** on an internal held-out set via a human-in-the-loop retraining loop.
- Orchestrated fault-tolerant training and serving workflows on **Temporal** and **SkyPilot** across **RunPod** and **Hugging Face Endpoints**, standardizing deploys across models.
- Benchmarked open **Vision-Language Models** (incl. Gemma 3) with a reproducible **ClearML** test suite, surfacing a Flash-Attention/SDPA correctness issue in the serving path; contributed to 314e's engineering write-ups on the AI-native [document-processing system](#) and [Dexit extraction pipeline](#).
- Cut model **cold-start latency** **77%** by pre-packaging weights and tokenizer into the runtime image and warming the serving process.
- Cut model training time from **90 hours to 25 hours** on >500 GB datasets via targeted hyperparameter tuning, and eliminated OOM errors with a custom memory-efficient iterable dataloader.
- Re-engineered single-node Python ETL into distributed **PySpark** jobs over **FHIR** resources for multiple clients, cutting data-loading time **~80%** and unblocking training on 500 GB+ datasets.

Banach Technologies

Machine Learning Intern

Singapore (Remote)

Apr 2024 – Jul 2024

- Engineered and optimized **high-frequency trading** strategies, reducing data-processing latency through targeted performance tuning.
- Built and deployed ML models from scratch against live market APIs to execute hedging strategies, backed by a comprehensive unit-test suite for core trading libraries.

Alemeno

Software / ML Intern

Maharashtra (Remote)

Jul 2023 – Apr 2024

- Built a data-preprocessing pipeline for **semantic-segmentation** models over large-scale GIS raster data, and implemented distributed computer-vision algorithms (Douglas–Peucker, Jarvis March).
- Deployed scalable **Django** apps on **AWS** with an end-to-end **CI/CD** pipeline, and hardened a high-performance raster-processing app through refactoring and unit tests.

Bewgle

Data Analyst

Bengaluru (Remote)

May 2022 – Jul 2022

- Applied **NLP** to proprietary Amazon-review datasets for product-trend insights; automated preprocessing in Python/Bash, cutting data redundancy by **90%**.

PROJECTS

[Resume Parser](#) — PyTorch, LLMs, HuggingFace

2024

- Built an end-to-end **LLM-powered** resume parser with a modular OOP codebase and a **Streamlit** UI producing structured JSON output.

[rumpy](#) — NumPy in pure Rust — Rust

2025

- Building a **zero-dependency** ndarray library in pure **Rust** (flat-buffer / shape / strides layout), reimplementing broadcasting, axis reductions, and cache-friendly (ikj) matmul from first principles; spec-driven with ~30 milestone tests benchmarked against NumPy.

PUBLICATIONS

[Medical Waste Classification using Deep Learning](#) — Verma M., Kumar A., Kumar S.

IEEE, IATMSI 2022

First author. Trained a CNN reaching **98.4%** accuracy; deployed as an interactive TensorFlow.js web app.

SKILLS

ML / LLM: PyTorch, Hugging Face, vLLM, SGLang, VLM evaluation & benchmarking, model fine-tuning, RAG, inference optimization

MLOps & Infra: Docker, Kubernetes, MLflow, ClearML, Temporal, SkyPilot, RunPod, FastAPI, PySpark, AWS SageMaker

Languages & Core: Python, C++, Rust, Bash; NumPy, Pandas, scikit-learn, PostgreSQL, Git, Linux

Open source: Contributor @ PyMC

EDUCATION

ABV-IIITM Gwalior

B.Tech, Computer Science — SGPA 8.67

Madhya Pradesh, India

Graduated 2024